

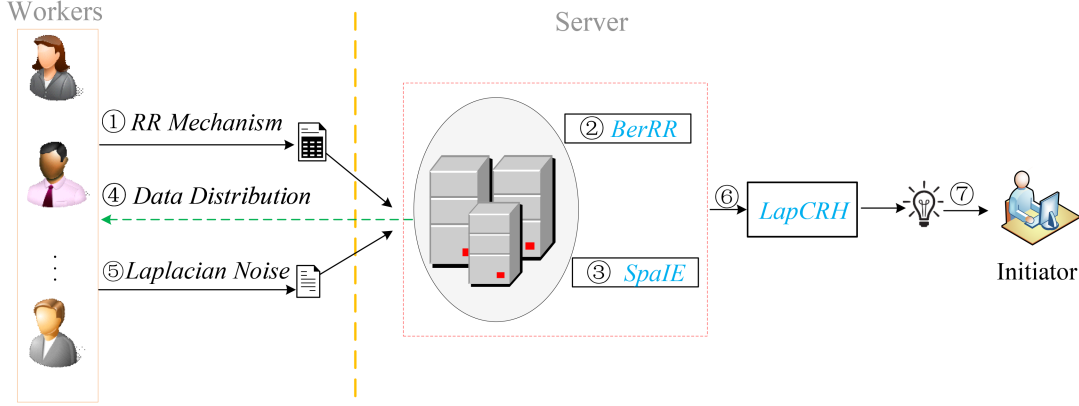


Fig. 1. Overview of OSCAR with their associated key techniques marked in blue.

both builds upon the privacy guarantees and data quality improvements of previous phases while preparing the groundwork for subsequent phases, creating a cohesive system that maintains both rigorous privacy protection and high data utility throughout the entire truth discovery process.

B. RR-Based Bernoulli Matrix Factorization

Bernoulli distribution is usually used to model binomial data, which can represent whether a worker participates in a task. Moreover, due to the strong power for high-dimensional data representation of matrix factorization, we first design *BerRR* to represent workers and tasks and alleviate noise from privacy protection under LDP, which incorporates *RR* into Bernoulli distribution with privacy parameter $\frac{\epsilon}{2}$. We solve the new objective function using *BCD* with the obtained gradients.

In *BerRR*, after obtaining the perturbation results using *RR*, we can calculate η_t , which represents the true proportion of workers who have completed the t -th task, using the formula $\frac{z_t - M_t q}{p - q}$ according to (7), where z_t denotes the noisy frequency of the t -th task, M_t is the number of workers completing the t -th task. $p = \frac{e^{\frac{\epsilon}{2}}}{1 + e^{\frac{\epsilon}{2}}}$, and $q = \frac{1}{1 + e^{\frac{\epsilon}{2}}}$ are the probabilities of answering true and false values respectively.

Consequently, we derive (8), which represents the proportions of workers who have completed p_t and not completed p_f the t -th task, respectively.

$$\begin{cases} p_t = \eta_t p + (1 - \eta_t)(1 - p) \\ p_f = (1 - \eta_t)p + \eta_t(1 - p) \end{cases} \quad (8)$$

where p_t represents the probability of actually completing the task η_t and still completing it after being perturbed p , as well as the probability of not completing the task $1 - \eta_t$ while being perturbed for completion $1 - p$. Similarly, we have p_f .

Due to the powerful representation ability of matrix factorization for sparse data, whether a worker participates in a task can clearly be modeled as a Bernoulli distribution, with (8), we have

$$\begin{aligned} p(I_{ij} | U_i, V_j) \\ = \begin{cases} \alpha g(U_i^T V_j) + (1 - \alpha) p_t, & I_{ij} = 1 \\ \alpha [1 - g(U_i^T V_j)] + (1 - \alpha) p_f, & I_{ij} = 0 \end{cases} \end{aligned} \quad (9)$$

where $g(x) = 1/(1 + e^{-x})$, I_{ij} indicates whether the i -th worker has done the j -th task, α is used to balance the representational power of matrix factorization and *RR*. Thus, we have

$$\begin{aligned} p(I | U, V) &= \prod_{i=1}^M \prod_{j=1}^N p(I_{ij} | U_i, V_j) \\ &= \left(\prod_{I_{ij}=1} (\alpha g(U_i^T V_j) + (1 - \alpha) p_1) \right) \\ &\quad \left(\prod_{I_{ij}=0} (\alpha [1 - g(U_i^T V_j)] + (1 - \alpha) p_2) \right) \end{aligned}$$

To produce a sparse effect, we let

$$\begin{cases} p(U | \sigma_U) = \prod_{i=1}^M L(U_i | 0, \sigma_U) \\ p(V | \sigma_V) = \prod_{j=1}^N L(V_j | 0, \sigma_V) \end{cases},$$

where $Lap(\cdot)$ denotes the Laplacian distribution.

Similar to (5), we have (10).

$$\arg \min_{U, V} \left\{ \begin{aligned} & - \sum_{I_{ij}=1} \ln (\alpha g(U_i^T V_j) + (1 - \alpha) p_1) \\ & - \sum_{I_{ij}=0} \ln (\alpha [1 - g(U_i^T V_j)] + (1 - \alpha) p_2) \\ & + \lambda \sqrt{\|U\|^2} + \lambda \sqrt{\|V\|^2} \end{aligned} \right\}. \quad (10)$$

The process begins by computing the gradients of U and V from (10). Subsequently, each worker uses half of their privacy budget to perturb their task participation status. These perturbed responses are then used to train the model using *SGD* until convergence.

C. Sparse Recovery via Implicit Enumeration

To tackle the dense concern and reconstruct the original K distribution, after receiving the perturbed high-dimensional representation U and V by *BerRR*, we develop a sparse structure recovery method *SpaIE* based on implicit enumeration to minimize the difference between the density obtained from *BerRR* and the true density estimated by *RR*.

In *SpaIE*, we formalize a 0-1 integer programming problem with L_1 norm in (11), where M and N are the number of workers and tasks respectively. Its physical significance lies in generating sparsity effect through $\tilde{I}$ and minimizing the difference $(\tilde{I} - g(UV))^2$ between the estimated distribution $\tilde{I}$ and the obtained high-dimensional representations, while minimizing the difference $(1^M \cdot \tilde{I} \cdot 1^N - \sum_{t=1}^N \eta_t)^2$ between $\|\tilde{I}\|_1$ and